

Cover Letter

Original submission to IEEE Access

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To: The Editor-in-Chief, *IEEE Access*

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Manuscript title: On the Waveform-to-Bit Gap in Single-Channel Blind Source Separation of Co-Frequency Signals: A Decision-Identical Evaluation Protocol and Controlled Diagnosis

Manuscript type: Original Research Article

Dear Editor,

We are pleased to submit the above manuscript for consideration in *IEEE Access*. The paper contributes *measurement infrastructure and a controlled diagnosis* for a question that the single-channel blind source separation (SC-BSS) literature has so far left untested: whether the waveform-fidelity gains that the field optimises and reports (SI-SDR/SDR) actually reach the demodulated bits of co-frequency communication signals. The work sits squarely within the journal's scope in signal processing, communications, and machine learning for the physical layer, and it is designed for reuse: the benchmark, the evaluation receiver, and the training-time proxy are released in full.

Key contributions.

1. **A decision-identical evaluation protocol.** All bit-level claims in the literature are only as strong as the receiver that produces them. We contribute (i) an offset-compensated receiver (oracle down-conversion, matched filtering, symbol-grid sampling, unit-power normalisation, phase-only alignment) producing compensated SER and Gray-mapped BER from shared decisions, and (ii) a differentiable soft-demodulation loss whose front-end is verified *symbol-for-symbol identical* to that evaluation receiver (matched filter agreeing with the reference implementation to a relative error of 3.5×10^{-7}). This removes the train/eval mismatch that has made earlier SER claims hard to compare, and it is directly reusable wherever separation is evaluated against downstream bits.
2. **A quantified waveform-to-bit mismatch with a located cause.** On a variable-count ($K \in \{1, 2, 3\}$) co-frequency benchmark at SIR ≈ 0 dB, a +4.1 dB SI-SDR improvement moves compensated SER only from 0.5049 to 0.5027—even +8.4 dB at -10 dB SNR buys nothing—and the flatness persists across an SIR sweep to 20 dB at fixed $K=2$. Per- K decomposition localises the damage: a single interferer raises the baseline SER $3.8\times$ (0.146 to 0.561), so the gap is interference-limited, not noise-limited. To our knowledge this mismatch has not previously been quantified under statistically controlled conditions.
3. **A sign-structured effect of the task-oriented loss.** The soft-demodulation term is practically negligible in the pool (≤ 0.03 pp in the fair additive comparison), but the pool hides a reproducible structure: the term *helps* at $K=2$ (5/5 seeds positive in all four contrasts) and *hurts* at $K=3$ (0/5 seeds)—the number of interferers, not the loss weight, sets the sign. The same term repairs occupancy gating (count accuracy $0.65 \rightarrow 0.85$, hallucination rate $0.15 \rightarrow 0.05$) and, deprived of the waveform anchor, collapses into reward hacking (SI-SDRi -9.4 dB), delimiting where such proxies can be trusted.

4. **A controlled attribution experiment and a concrete prescription.** A joint soft-information head that bypasses explicit separation fails at chance; controlled probes (same head, same data, oracle vs. blind synchronisation as the only variable) attribute the failure to burst-level blind carrier/phase recovery, and at $K=2$ even a frequency oracle fails under phase-entangled interference. The practical conclusion for the co-frequency and PCMA communities is direct: closing the waveform-to-bit gap requires explicit synchronisation, equalisation, and joint-detection structure—not a re-weighted loss.

Statistical standard. All headline numbers are multi-seed (five seeds). Contrasts use an exact sign-flip permutation test at the seed level and a Monte-Carlo permutation test at the cell level, with Bonferroni control over eight pre-declared contrasts; the $n=5$ power ceiling ($p \geq 0.0625$ attainable) is stated explicitly rather than smoothed over, and a selection-bias artefact in a baseline architecture is identified and corrected with a matched-pair control.

Reproducibility. All training, evaluation, analysis, and figure/table-generation code, including the synthetic data generator and the deterministic test grid (test seed 99999), is publicly available at https://github.com/BinNong/single_channel_source_seperation/tree/main/paper5_task_oriented; no external datasets are required. The exact command behind every number is recorded in the repository’s experiment log.

Related submissions. Two related companion manuscripts are under review elsewhere and share no results with the present paper: a closed-set separator design (the complex CNN with squeeze-and-excitation used as a baseline family here) at *Wireless Personal Communications* (Springer), and the open-world variable- K study that this paper’s benchmark and slot separator build upon at *China Communications*. The present manuscript is fully self-contained; the cited companion manuscript (currently listed as “under review”) is available to the editor and reviewers on request.

Originality and ethics.

- The manuscript is original, has not been published elsewhere, and is not under consideration by any other journal.
- All authors have approved the submission; the authors have no conflicts of interest to declare.
- No human subjects or animal experiments are involved; all experiments use synthetic signals from our released generator.

Author contributions. B. Nong conceived the study and the task-oriented formulation, ran the multi-seed experiments and statistical analyses, and drafted the manuscript. W. Fu contributed to the conceptualisation of the evaluation framework. Z. Jiang contributed resources and validation. All authors reviewed and approved the final manuscript.

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Suggested reviewers. We have no specific reviewer suggestions; we trust the editorial board to identify expert reviewers in co-frequency interference management and machine-learning-based physical-layer signal processing.

Thank you for considering our manuscript. We look forward to hearing from you.

Sincerely,

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